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4.8 Space Physics

AQA GCSE Physics (8463) — Paper 2 (Triple / Separate Physics)

4.8.1 Our Solar System

The **Solar System** consists of the Sun (a star at the centre), eight planets orbiting the Sun, moons orbiting the planets, dwarf planets, asteroids, and comets. Everything orbits due to the Sun's gravitational attraction.

The Eight Planets

The eight planets, in order from the Sun, are:

1. **Mercury**
2. **Venus**
3. **Earth**
4. **Mars**
5. **Jupiter**
6. **Saturn**
7. **Uranus**
8. **Neptune**

Mnemonic: "My Very Educated Mother Just Served Us Noodles"

Two Categories of Planets

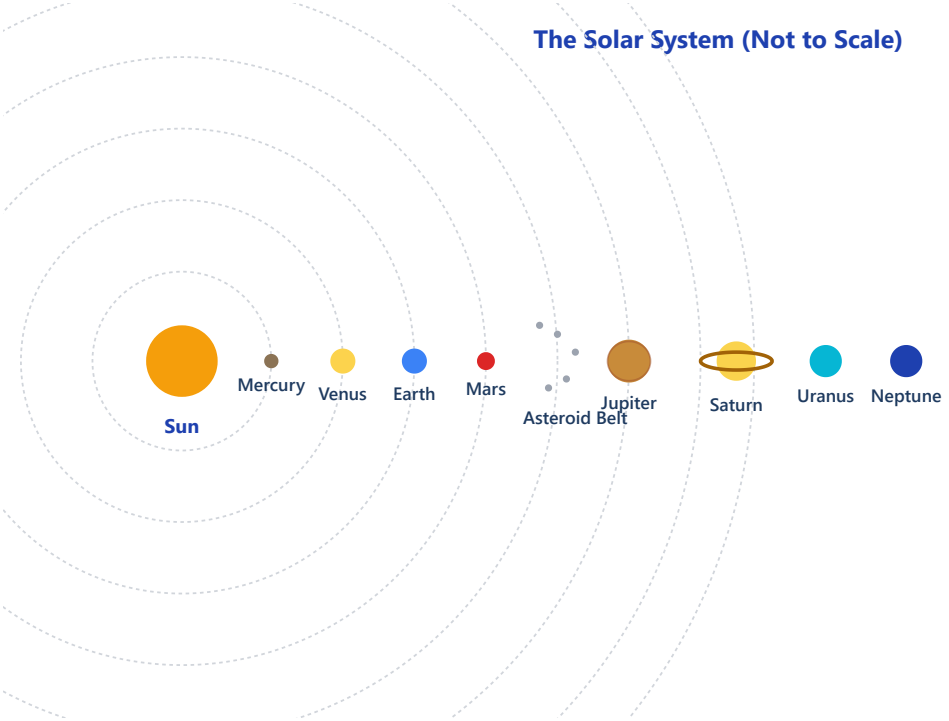
Terrestrial (Rocky) Planets	Gas Giants

Mercury, Venus, Earth, Mars	Jupiter, Saturn, Uranus, Neptune
Small to medium size	Very large size
High density (solid surface)	Low density (gaseous)
Located in inner solar system	Located in outer solar system
Few or no moons	Many moons (Jupiter has 95+, Saturn has 146+)

A **dwarf planet** is a celestial body that orbits the Sun and is massive enough to be round, but has not cleared its orbital path of other objects. Examples: Pluto, Ceres, Eris.

Natural satellites (moons) are large objects that orbit planets (e.g. Earth's Moon). **Artificial satellites** are human-made objects that orbit planets or stars (e.g. GPS satellites, the International Space Station, communication satellites).

The Solar System (Not to Scale)



Key Features:

- Terrestrial: Mercury–Mars
- Gas giants: Jupiter–Neptune
- Asteroid belt between Mars and Jupiter
- All orbits in same plane
- Sun at the centre
- Gravity holds all orbits
- Not drawn to scale

Other Solar System Objects

Asteroid Belt: Located between Mars and Jupiter, contains thousands of rocky fragments (leftover from solar system formation).

Comets: Icy bodies with very elongated elliptical orbits. When near the Sun, solar radiation vaporizes ice, forming a bright tail that always points away from the Sun.

> **Exam Tip:** Learn the planet order using the mnemonic. Know that terrestrial planets are inner and small, while gas giants are outer and large. Be able to identify the asteroid belt location and explain dwarf planets (not clearing their orbit).

Our Place in the Universe

> The **Milky Way** is a galaxy containing approximately 100 to 200 billion stars, including our Sun. Our Solar System is located in one of the spiral arms. The universe contains hundreds of billions of galaxies.

4.8.2 Formation of the Sun and Solar System

> The solar system formed approximately **4.6 billion years ago** from a giant cloud of dust and gas (a **nebula**).

How the Solar System Formed

The process:

1. **Gravitational collapse:** The nebula began to collapse under its own gravity, pulling all material inward.
2. **Formation of the Sun:** Most of the mass accumulated at the centre, forming a hot, dense ball. Temperature and pressure increased until hydrogen fusion began, igniting the Sun.
3. **Formation of planets:** Surrounding the new Sun, dust and gas particles collided and stuck together (accretion), gradually building up into planets.

4. **Planetary clearing:** As planets formed, they cleared their orbital zones of other debris (through collisions and gravitational interactions).

> The Sun is now in its **main sequence** phase — a stable period lasting approximately 10 billion years where the Sun fuses hydrogen into helium in its core. About 4.6 billion years have elapsed; approximately 5.4 billion years remain.

> **Exam Tip:** Remember: nebula → collapse → Sun ignites → planets form via accretion. The Sun's age (4.6 Ga) and the solar system's formation can be dated using radioactive decay of isotopes in meteorites.

4.8.3 Life Cycle of a Star

> All stars begin their lives as a **nebula** — a cloud of dust and gas in space. Over millions of years, gravity pulls this cloud together, eventually forming a star.

Stages of Star Evolution

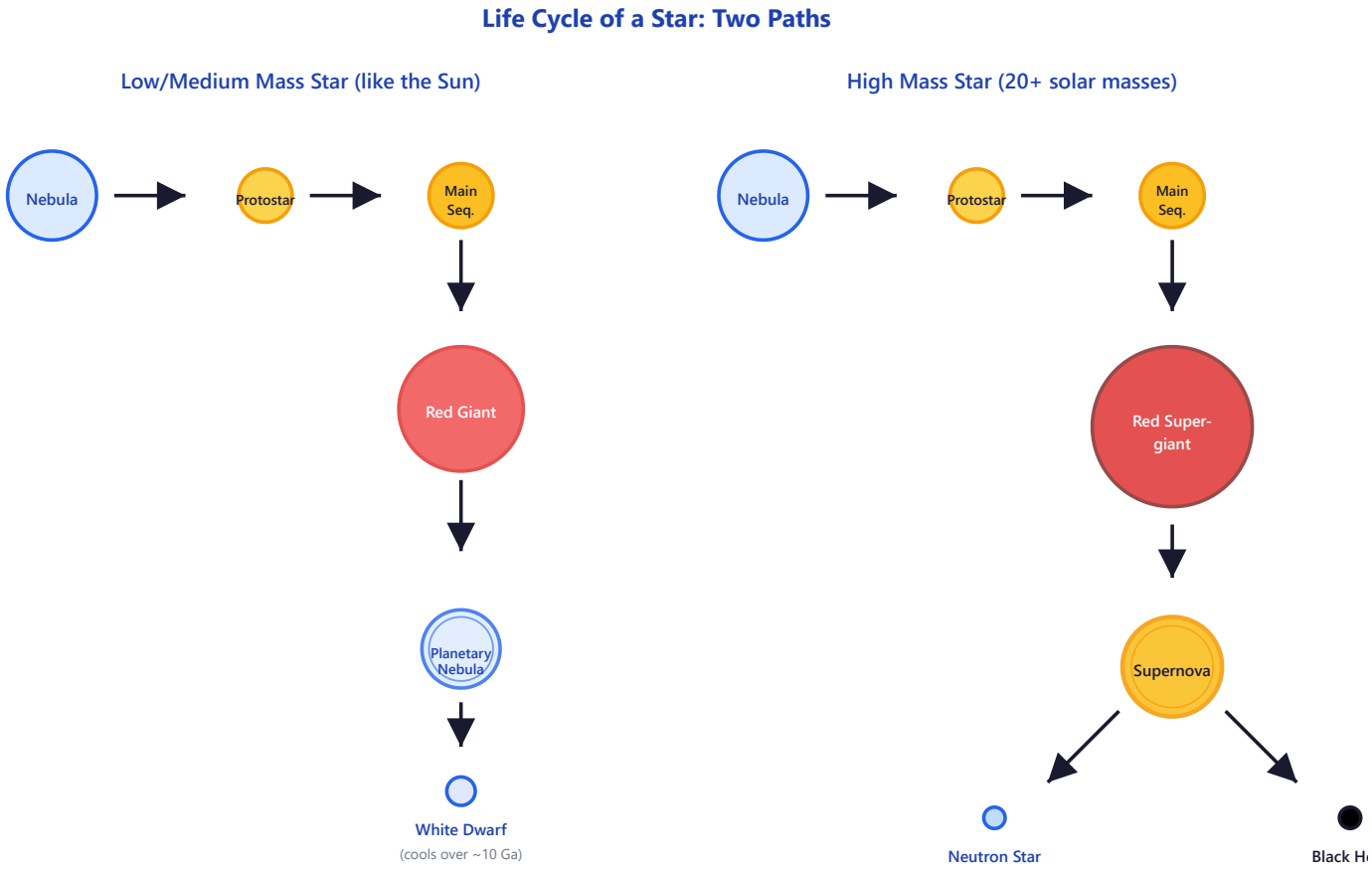
Stage 1: Nebula and Protostar

- A giant cloud of dust and gas collapses under gravity.
- As it collapses, gravitational potential energy converts to heat.
- A hot, dense centre forms — the **protostar**.
- Temperature rises until the core reaches ~10 million K.

Stage 2: Main Sequence

- Hydrogen fusion begins when core temperature exceeds 10 million K.
- Gravity (pulling inward) balances thermal pressure (pushing outward) — **hydrostatic equilibrium**.
- The star is stable and shines by converting hydrogen to helium.
- This phase lasts the majority of the star's lifetime (for Sun: ~10 billion years total, 4.6 Ga so far).

Stage 3: End of Life — Two Paths Depending on Mass



Low/Medium-Mass Stars (Like Our Sun)

1. Red Giant: When hydrogen in the core is exhausted, the star expands dramatically and cools (surface temperature drops). The core, now hotter, fuses helium into carbon and oxygen. The star glows red.

2. Planetary Nebula: The star sheds its outer layers into space, creating a beautiful nebula of ejected gas.

3. White Dwarf: The remaining core, about the size of Earth, is incredibly dense. It slowly cools over billions of years.

4. Black Dwarf: Theoretical end state — a white dwarf that has cooled completely. The universe is not yet old enough for this to occur (universe is 13.8 Ga old).

High-Mass Stars (More Than ~20 Solar Masses)

1. Red Supergiant: Similar to red giant but much larger and more luminous. The core fuses heavier elements: helium → carbon, carbon → neon, neon → magnesium, etc., up to iron.

2. Supernova: Iron fusion absorbs energy rather than releasing it. The core can no longer support itself and collapses catastrophically, then rebounds in a violent explosion — a supernova. For a brief moment, a supernova can outshine an entire galaxy.

3. Remnant — Two Possibilities:

- **Neutron star:** The crushed core (neutrons packed together) about 20 km across, incredibly dense, and often rotating rapidly (pulsar).
- **Black hole:** For the most massive stars, gravity is so strong that nothing — not even light — can escape once it crosses the event horizon.

> **Nucleosynthesis:** Heavy elements (metals) are created through nuclear fusion in stars. Fusion inside stars creates all elements up to iron (Fe). Elements heavier than iron (gold, silver, uranium, etc.) are created only during supernova explosions. The supernova scatters these heavy elements across space, which eventually become part of new solar systems and planets.

Star Type	Main Sequence	End of Life	Final Remnant
Low/medium mass (like Sun)	~10 billion years	Red giant → Planetary nebula	White dwarf → Black dwarf
High mass (>20 M_sun)	~10 million years	Red supergiant → Supernova	Neutron star or Black hole

> **Exam Tip:** You are made of stardust! Carbon, oxygen, nitrogen, iron, and other atoms in your body were formed in stars or supernovae billions of years ago. High-mass stars produce elements heavier than iron only during supernovae, which is why these elements are rare on Earth.

4.8.4 Orbital Motion, Satellites, and Resonance

Orbital motion occurs when an object moves in a closed path (orbit) around another object due to gravitational force. The gravitational force is always directed toward the centre of the orbit and provides the **centripetal force** needed to keep the object moving in a circle (or ellipse).

Key Principles of Orbits

Gravitational Force as Centripetal Force:

- The Sun's gravity pulls planets toward the Sun (centripetal direction).
- This constant inward force causes the planet's velocity direction to change continuously → the planet orbits rather than flying off in a straight line.
- The planet is constantly accelerating (direction changes), but speed may be constant for a circular orbit.

Orbital Speed and Altitude:

- **Lower orbits (closer to Earth):** Stronger gravitational pull → need higher orbital speed to stay in orbit.
- **Higher orbits (farther from Earth):** Weaker gravitational pull → need lower orbital speed.
- At any given orbital radius, there is only one orbital speed that maintains a stable circular orbit.

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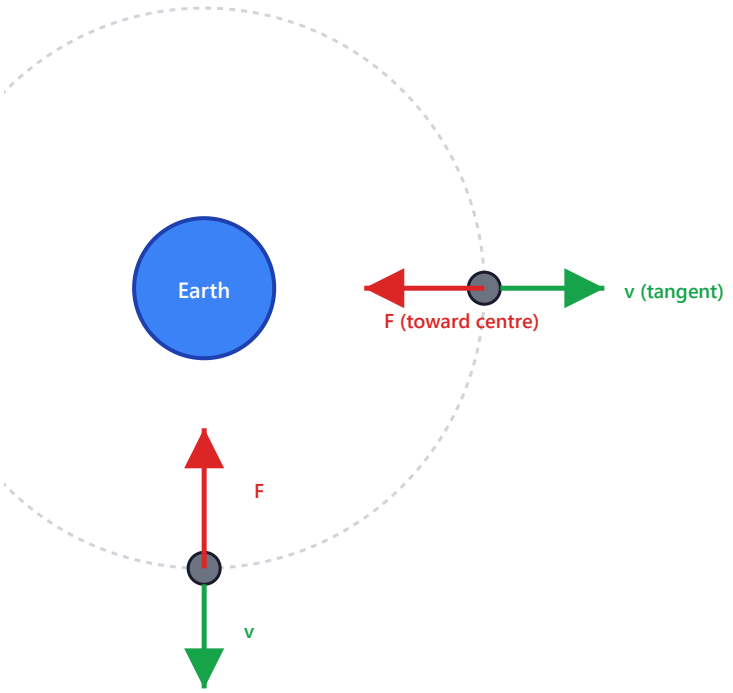
Orbital Motion: Velocity and Gravitational Force



Key Points:

- Satellite always moves tangent to its orbit
- Gravitational force always points toward Earth's centre
- This inward force continuously changes the velocity direction
- Lower orbit → faster
Higher orbit → slower
- Constant acceleration even at constant speed (direction always changes)

Types of Satellites

Geostationary Satellites:

- Orbital altitude: approximately 36,000 km above Earth's surface.
- Orbital period: exactly 24 hours (same as Earth's rotation).
- Remains stationary above a fixed point on Earth's equator.
- Uses: TV and communication broadcasts, weather monitoring.

Low Earth Orbit (LEO) Satellites:

- Orbital altitude: typically 400–2000 km above Earth's surface.
- Orbital period: 90 minutes to a few hours (fast orbit).
- Moves relative to Earth; passes over different locations.
- Examples: International Space Station, Earth observation satellites, GPS satellites.
- Uses: Earth imaging, climate monitoring, communications (polar regions).

> A **geostationary satellite** has an orbital period equal to Earth's rotational period (24 hours), keeping it stationary above one point on the equator. **Low Earth Orbit (LEO) satellites** orbit

much closer and faster, completing an orbit in ~90 minutes.

Exam Tip: Geostationary satellites stay in the same spot in the sky. LEO satellites move quickly across the sky. Higher orbits have lower speeds; lower orbits have higher speeds. Remember: orbital period increases with orbital radius.

4.8.5 Red-Shift and the Doppler Effect

Red-shift: When a source of light moves away from an observer, the observed wavelength increases (shifts toward the red, longer-wavelength end of the spectrum). When moving toward the observer, wavelength decreases (blue-shift).

The Doppler Effect for Light

Source moving away: Light waves are stretched to longer wavelengths → red-shift (lower frequency, lower energy).

Source moving toward: Light waves are compressed to shorter wavelengths → blue-shift (higher frequency, higher energy).

Source stationary: No shift in wavelength.

Observing Red-Shift from Galaxies

Key observation: Spectral absorption lines from distant galaxies are shifted toward the red end of the spectrum compared to their laboratory wavelengths.

- Nearly all distant galaxies show red-shift.
- More distant galaxies have larger red-shifts.
- This indicates galaxies are moving away from us.
- More distant = faster recession speed.

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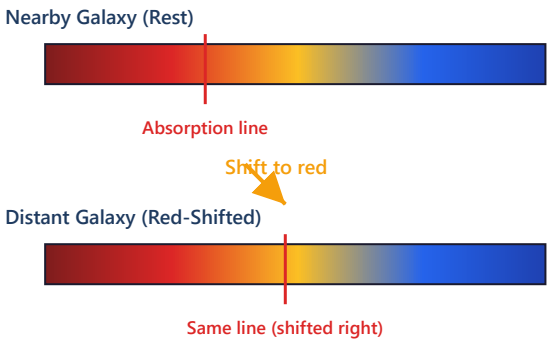
Red-Shift in Distant Galaxies



What Red-Shift Tells Us:

- Wavelength longer than laboratory value
- Galaxy is moving away from us (Doppler effect)
- Larger red-shift = faster recession speed
- More distant galaxies recede faster
- Evidence: Universe is expanding

Exam Tip: Red-shift = galaxy moving away. Blue-shift = approaching (rare for galaxies). The Doppler effect applies to light waves just as it does to sound (ambulance siren). Measuring red-shift allows us to determine recession speeds of distant galaxies.

4.8.6 The Big Bang Theory and the Expanding Universe

The **Big Bang theory** states that the universe began as an extremely hot, dense point (a singularity) approximately **13.8 billion years ago** and has been expanding ever since.

Evidence for the Big Bang Theory

Evidence 1: Red-Shift (Hubble Expansion)

- All distant galaxies are red-shifted, indicating they move away from us.
- More distant galaxies have larger red-shifts (faster recession).
- This pattern is consistent with an expanding universe originating from a single point.
- If we "reverse time," all galaxies converge to a common origin point — the Big Bang.

Evidence 2: Cosmic Microwave Background Radiation (CMBR)

- Astronomers detect a uniform microwave radiation filling the entire universe.
- This is the "leftover" radiation from the extremely hot early universe, now cooled to about 2.7 K.

- Predicted by Big Bang theory; discovered in 1964.
- The uniformity and temperature of CMBR match Big Bang predictions remarkably well.

> The **Cosmic Microwave Background Radiation (CMBR)** is the thermal radiation left over from the early, hot universe. It fills all of space uniformly and is evidence that the universe was once in an extremely hot, dense state.

Dark Matter and Dark Energy

Dark Matter: Observations show that galaxies rotate too fast to be held together by visible matter alone. Calculations indicate that approximately 85% of matter is "dark" (non-luminous). Its gravitational effects are observed, but its composition remains unknown.

Dark Energy: Since the 1990s, astronomers have observed that the expansion of the universe is accelerating (not decelerating as gravity would suggest). An unknown form of "dark energy" (about 68% of the universe's total energy density) appears to be driving this acceleration. Its nature remains one of cosmology's greatest mysteries.

> **Dark matter** (~27% of universe) — detected by gravitational effects but not visible. **Dark energy** (~68% of universe) — causes accelerating expansion; nature unknown. Ordinary visible matter comprises only ~5% of the universe.

Timeline of the Universe

Timeline of the Universe

Key Events:

- Big Bang: Universe began at a singularity with infinite density and temperature.
- Cosmic inflation: Universe expanded exponentially in first fractions of a second.
- First atoms: Protons and electrons combined when universe cooled to ~3000 K (380,000 years after Big Bang).
- Star and galaxy formation: Gravity pulled together hydrogen gas to form first stars and galaxies (100 million years onward).

Open Questions

What caused the Big Bang? We do not know. Physics breaks down at the singularity; quantum gravity is not yet understood.

What is dark matter? Candidates include WIMPs (Weakly Interacting Massive Particles), axions, and primordial black holes, but no detection has been confirmed.

What is dark energy? Possibly a cosmological constant (Einstein's "lambda"), vacuum energy, or a new field. Its nature remains a profound mystery.

What is the fate of the universe? If dark energy continues to dominate, the universe will expand forever, cooling toward absolute zero — a "heat death." But we cannot rule out surprises.

> **Exam Tip:** Big Bang evidence: (1) red-shift (galaxies receding), (2) CMBR (leftover heat from early universe). Be able to explain why these support Big Bang. Know the timeline: Big Bang 13.8 Ga → first atoms → first stars → galaxies → today. Dark matter and energy are real observations, not fictional.

Summary Table: Evidence for Big Bang Theory

Evidence	Observation	What It Tells Us

Red-Shift of distant galaxies	All distant galaxies show red-shift; more distant = larger red-shift	Galaxies recede; universe expands from a single point
Cosmic Microwave Background Radiation	Uniform microwave radiation at ~2.7 K throughout space	Leftover radiation from hot early universe; cooled as universe expanded
Abundance of light elements	Universe contains ~75% hydrogen, ~25% helium, trace heavier elements	Consistent with nuclear synthesis in first 3 minutes after Big Bang
Galaxy formation	First galaxies appeared ~100 million years after Big Bang	Gravity pulled together hydrogen to form galaxies from density fluctuations

Summary Table: Star Formation and Evolution

Stage	Low/Medium Mass Star	High Mass Star
1. Nebula	Giant cloud of gas and dust; collapses under gravity	
2. Protostar	Hot, dense centre forms; temperature rises	
3. Main Sequence	Hydrogen fuses to helium (~10 Ga)	Hydrogen fuses to helium (~10 Ma)
4. Post-Main Sequence	Red giant; helium fuses to C, O	Red supergiant; fuses up to iron
5. Ejection	Planetary nebula (gentle)	Supernova (violent explosion)
6. Remnant	White dwarf → Black dwarf (after 10+ Ga)	Neutron star or Black hole

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Summary Table: Satellites and Orbits

Property	Geostationary Satellite	Low Earth Orbit (LEO)
Altitude	~36,000 km	400–2,000 km
Orbital Period	24 hours (1 day)	~90 minutes
Orbital Speed	~3.1 km/s (slow)	~7.8 km/s (fast)
Position Relative to Earth	Stationary above one point on equator	Moves rapidly; passes over different locations
Uses	TV broadcasts, communications, weather	Earth observation, ISS, GPS, climate monitoring

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